PHISHING EMAIL DETECTION & AWARENESS REPORT

A Comprehensive Analysis of Real-World Phishing Attacks
with Technical Evidence and Business Prevention Guidelines

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1. Executive Summary

Phishing remains one of the most prevalent and damaging cyber threats facing businesses today. Over 80% of organisations experienced phishing attacks in recent years, with the average cost of a successful breach exceeding $4.9 million USD. Attackers continuously evolve their techniques, impersonating trusted brands such as PayPal, postal services, social media platforms, and major financial institutions to steal credentials, commit fraud, and gain unauthorised access to corporate systems.

This report presents a detailed technical and behavioural analysis of five real-world phishing email samples spanning five distinct attack categories: financial fraud, delivery scams, social media account takeover, IT/corporate credential theft, and banking fraud. Each sample was examined using professional security tools including Google Admin Toolbox Message Header Analyzer and MXToolbox SuperTool, with domain blacklist verification performed against 61–70 known threat databases.

Key findings confirm that all five samples failed standard email authentication checks (SPF, DKIM, DMARC), two sending servers were found on international spam blacklists, and multiple samples shared attack infrastructure — indicating coordinated phishing campaigns. This report concludes with actionable prevention guidelines designed to protect employees and reduce organisational risk.

2. Scope & Methodology

This assessment was conducted under a strictly read-only, passive analysis scope. No exploitation, login attempts, link activation, or any form of active attack was performed at any stage. All analysis was carried out on publicly available phishing samples collected from verified educational repositories and visual demonstration platforms.

2.1 What Was Analysed

- Sender domain authenticity and spoofing indicators
- Email authentication records: SPF, DKIM, and DMARC
- Email routing and server path analysis via full header inspection
- Domain reputation and blacklist status across 61–70 international threat databases
- Social engineering tactics: urgency, fear, impersonation, and psychological manipulation
- Attack chain mapping: email body → malicious link/button → credential harvest page

3. Tools Used

Tool	Purpose	URL
Google Admin Toolbox	Parse raw email headers, extract SPF/DKIM/DMARC, trace routing	toolbox.googleapps.com
MXToolbox SuperTool	Domain MX lookup, DMARC check, DNS verification	mxtoolbox.com/SuperTool.aspx
MXToolbox Blacklist	Check sending IP/domain against 61–70 spam blacklists	mxtoolbox.com/blacklists.aspx
Browser DevTools	Inspect link targets, redirect chains, phishing login pages	Built-in (F12)
Canva	Annotate phishing email screenshots with red-flag indicators	canva.com

GitHub – phishing_pot	Source of real-world .eml phishing samples	github.com/rf-peixoto/phishing_pot

4. Email Sample Analysis

4.1 Sample 1 — Financial Phishing (PayPal)

Category	Financial / Credential Theft
Impersonates	PayPal Inc.
Sender	support@paypaypal[.]net
Real Domain	paypal.com
Target	john[.]doe@mybusiness[.]com
Objective	Steal PayPal login credentials via a fake verification page

RISK LEVEL	■ HIGH

Annotated Email Screenshots

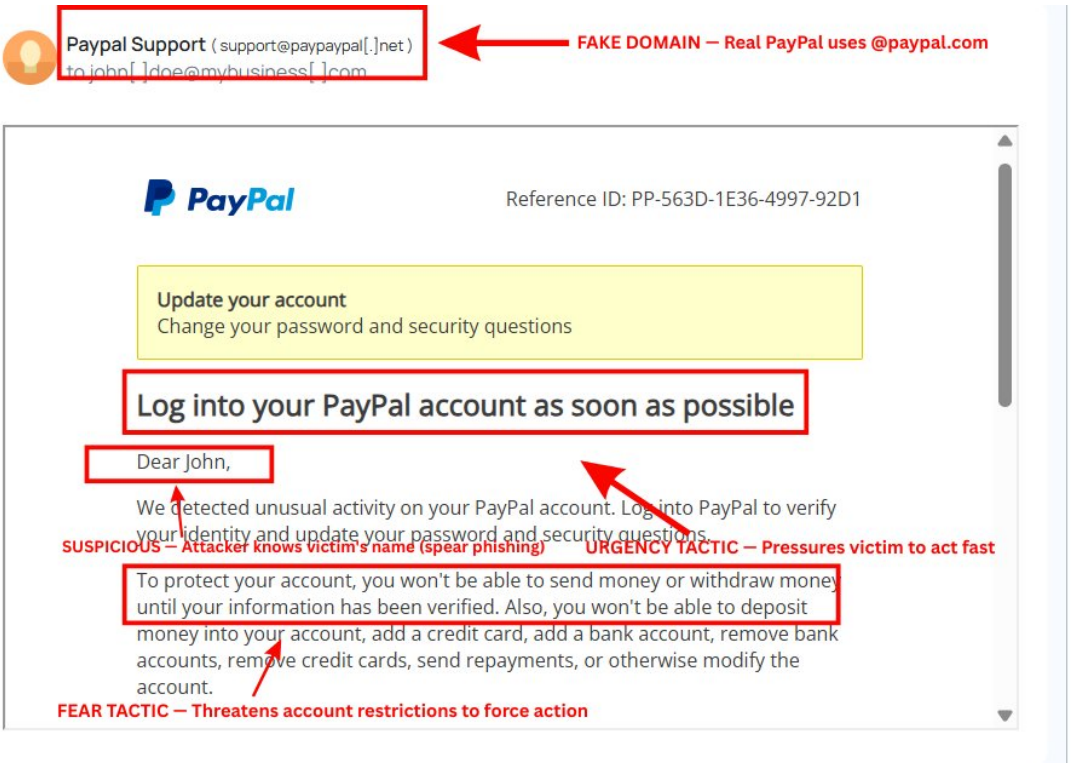

Figure 1a: PayPal phishing email — Part 1 (annotated: fake domain, urgency tactic, spear phishing indicator)

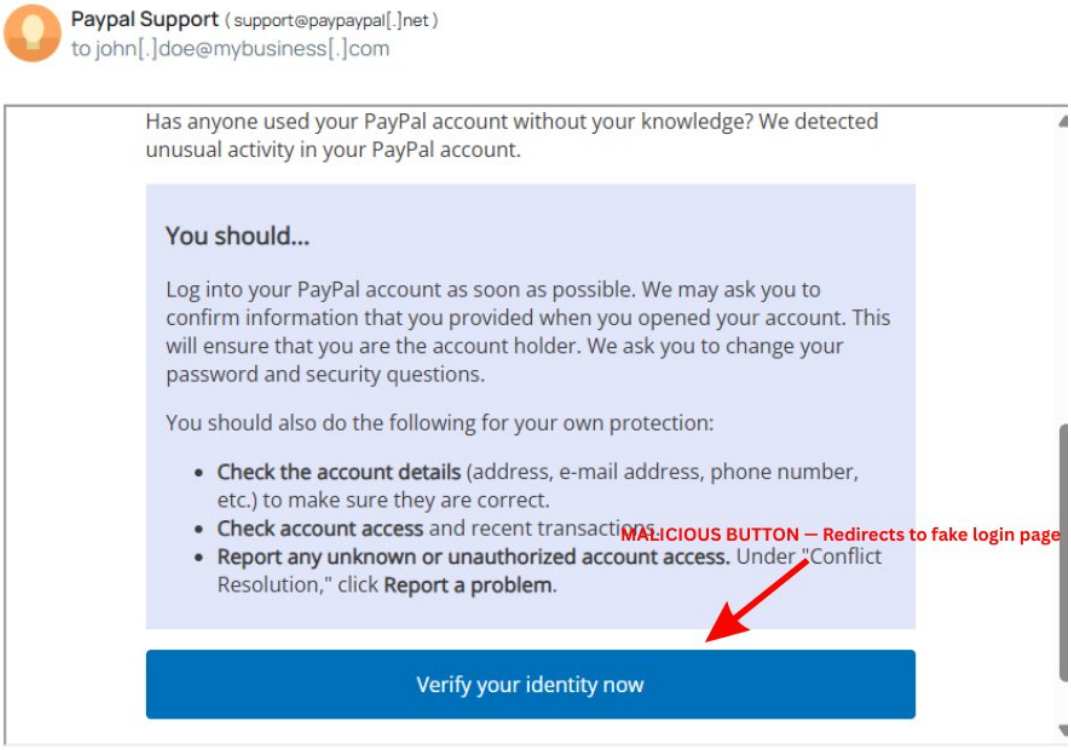

Figure 1b: PayPal phishing email — Part 2 (annotated: fear tactic and malicious verify button)

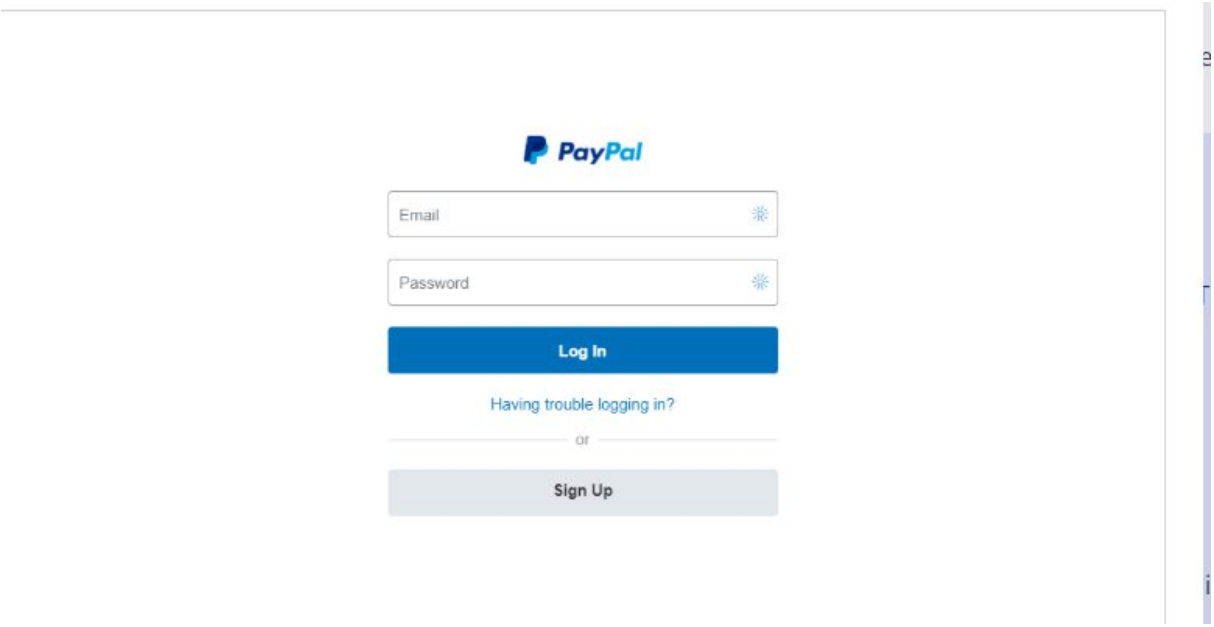

Figure 1c: Fake PayPal login page — destination after clicking 'Verify Now'

Red Flags Identified

■ **Fake Sender Domain:** support@paypaypal[.]net — Real PayPal uses @paypal.com only. The domain 'paypaypal' adds an extra 'pay' to deceive victims.

■ **Urgency Language:** 'Log into your PayPal account as soon as possible' — Creates time pressure to prevent rational thinking.

■ **Spear Phishing:** Uses victim's real name 'Dear John' — attacker had prior knowledge of the target.

■ **Fear Tactic:** Threatens account restrictions on sending, withdrawing, and depositing money to force immediate action.

■ **Fake Login Page:** Clicking 'Verify Now' redirects to a cloned PayPal login page to capture credentials.

■ **No Authentication:** Domain paypaypal.net has no SPF, DKIM, or DMARC records — confirmed via MXToolbox.

Attack Flow

Victim receives email from paypaypal[.]net → Reads urgency message → Alarmed by account restriction threats → Clicks 'Verify Now' → Redirected to cloned PayPal login page → Enters credentials → Attacker gains full account access → Financial loss.

4.2 Sample 2 — Delivery Phishing (Canada Post)

Category	Delivery Scam / Credential Theft
Impersonates	Canada Post
Sender	canada-post@webnotifications[.]net
Real Domain	canadapost.ca
Target	john[.]doe@mybusiness[.]com
Objective	Steal login credentials via a fake Canada Post portal

RISK LEVEL	■ HIGH

Annotated Email Screenshots

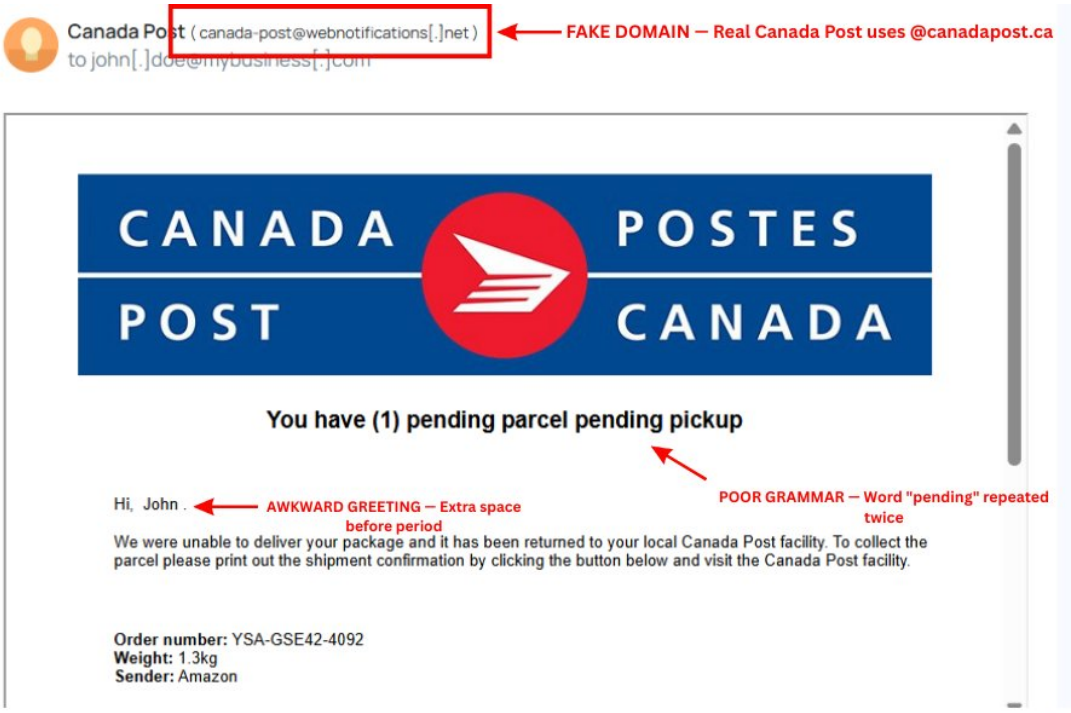

Figure 2a: Canada Post phishing email — annotated with red-flag indicators

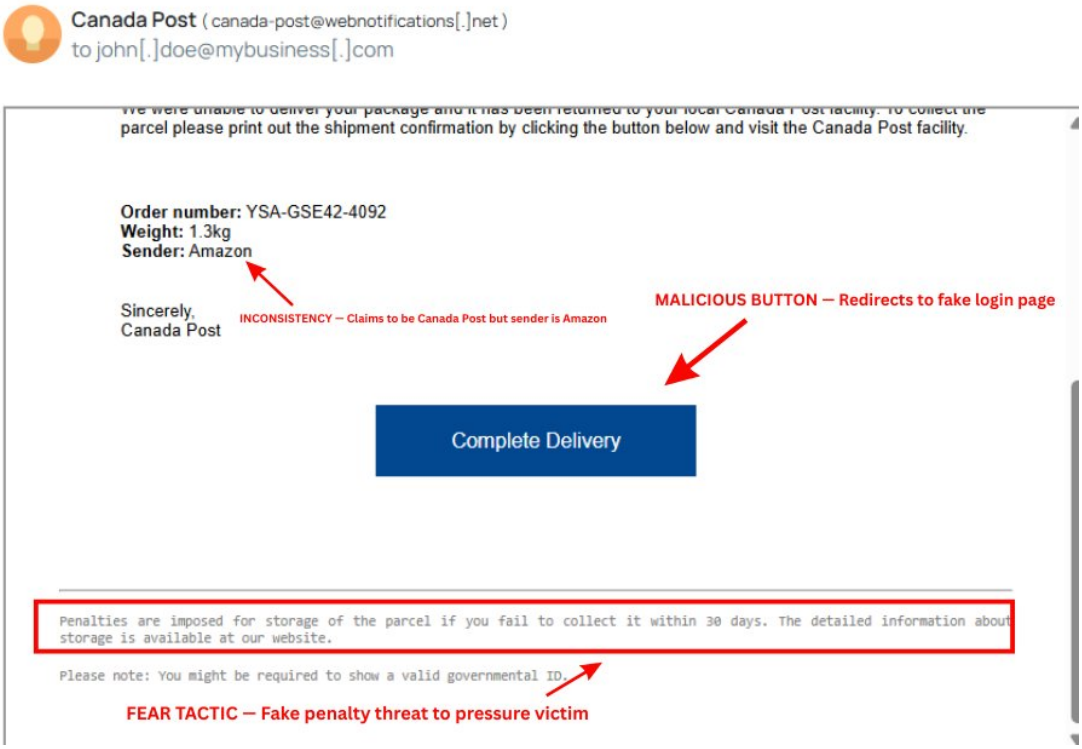

Figure 2b: Canada Post phishing email Part 2 — fear tactic and malicious delivery button

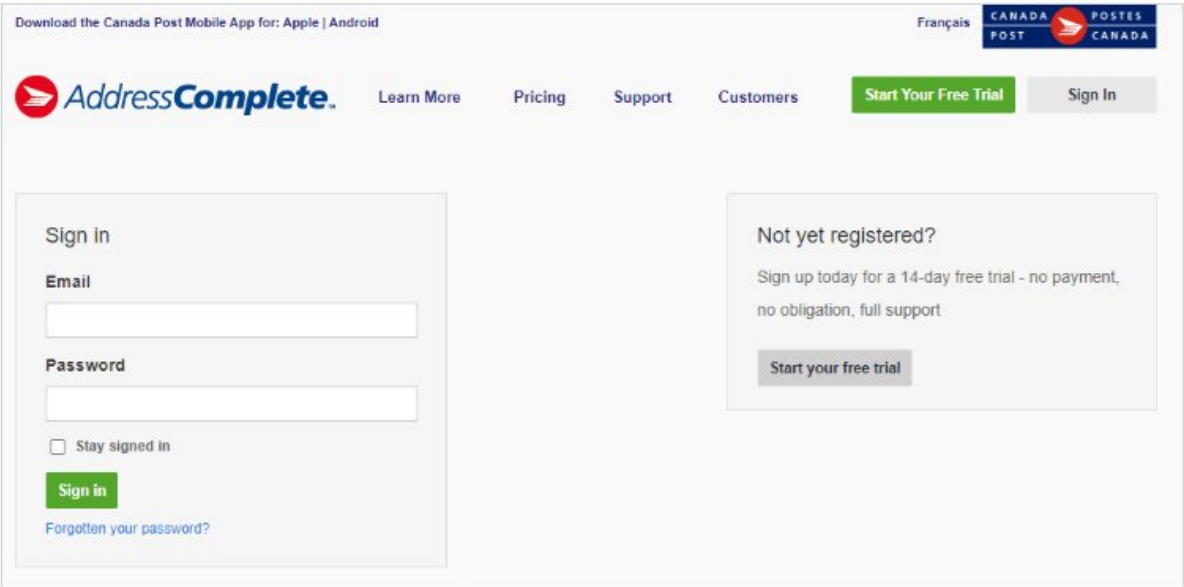

Figure 2c: Fake Canada Post login page — destination after clicking 'Complete Delivery'

Red Flags Identified

- **Fake Sender Domain:** canada-post@webnotifications[.]net — Completely unrelated to Canada Post's official domain canadapost.ca.
- **Grammar Error:** 'You have (1) pending parcel pending pickup' — Word 'pending' is repeated twice. Legitimate companies proofread communications.
- **Awkward Greeting:** 'Hi, John .' — Extra space before the period is a sign of automated phishing template generation.
- **Sender Inconsistency:** Claims to be Canada Post but states 'Sender: Amazon' — a logical contradiction exposing the fake.
- **Fake Penalty Threat:** 'Penalties are imposed for storage of the parcel if you fail to collect within 30 days' — False threat to create urgency.

- **Malicious Button:** 'Complete Delivery' button redirects to a third-party fake login portal, not the real Canada Post website.

Attack Flow

Victim receives delivery notification from webnotifications[.]net → Believes parcel is waiting → Feels urgency from 30-day penalty threat → Clicks 'Complete Delivery' → Redirected to fake login portal → Enters credentials → Attacker steals account access.

4.3 Sample 3 — Social Media Phishing (Instagram)

Category	Social Media / Account Takeover
Impersonates	Instagram (Meta)
Sender	instagram@webnotifications[.]net
Real Domain	instagram.com / facebookmail.com
Target	john[.]doe@mybusiness[.]com
Objective	Steal Instagram credentials via a fake security alert

RISK LEVEL	■ HIGH

Annotated Email Screenshots

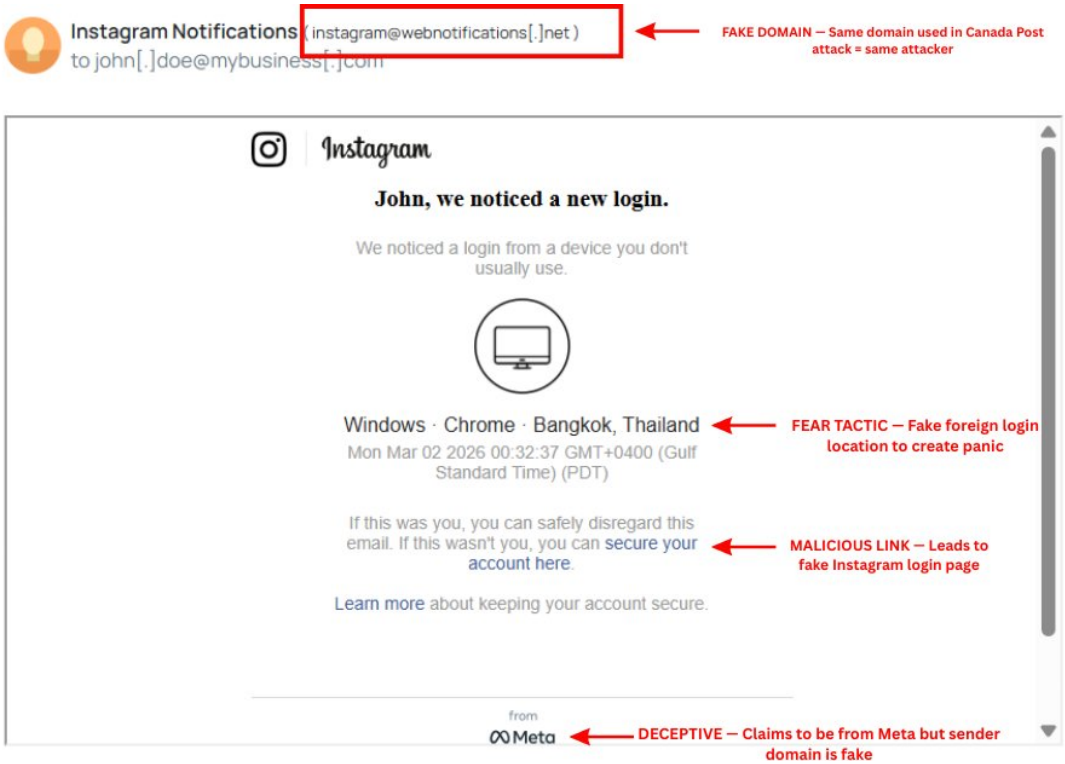

Figure 3a: Instagram phishing email — annotated with red-flag indicators

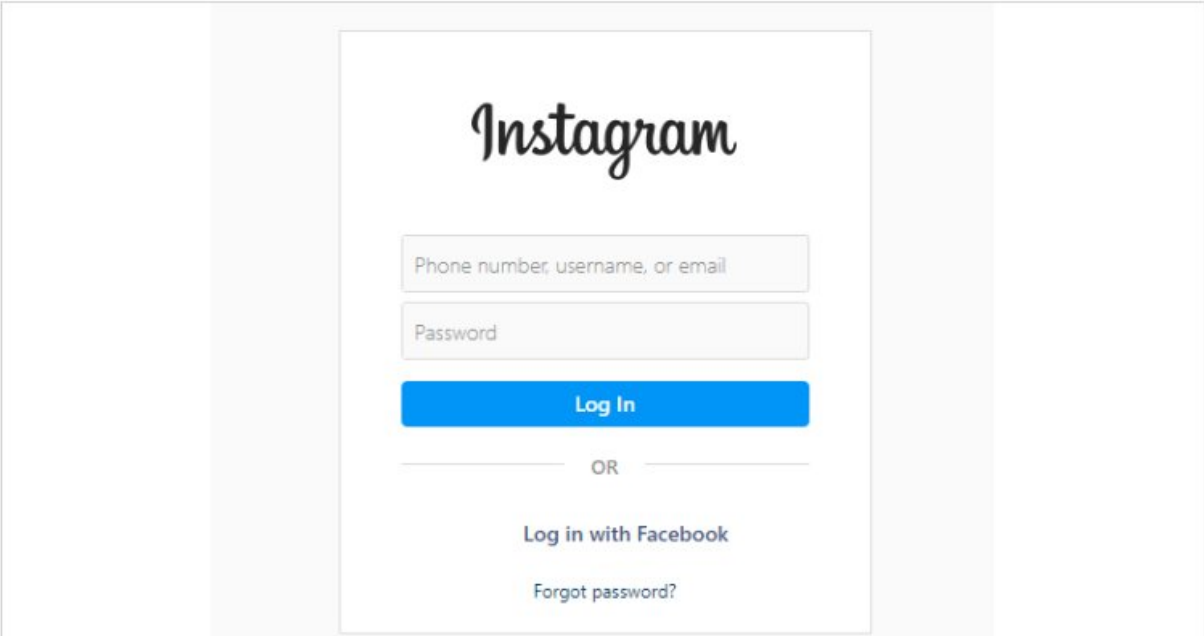

Figure 3b: Fake Instagram login page — destination after clicking 'secure your account here'

Red Flags Identified

- **Fake Sender Domain:** instagram@webnotifications[.]net — Same domain as Canada Post attack, confirming same attacker behind both campaigns.
- **Fear Tactic:** Claims a login from Bangkok, Thailand on an unknown device — designed to create immediate panic.
- **Malicious Link:** 'secure your account here' leads to a fake Instagram login page to harvest credentials.
- **False Meta Branding:** Email footer displays the Meta logo to appear legitimate, but sender domain has no affiliation with Meta Inc.
- **Spear Phishing:** Uses victim's real name 'John' — indicates attacker had prior knowledge of the target.

Attack Flow

Victim receives security alert claiming unusual login from Bangkok → Panic about account breach → Clicks 'secure your account here' → Redirected to cloned Instagram login page → Enters credentials → Attacker takes over account → Business Instagram page hijacked.

4.4 Sample 4 — IT/Corporate Phishing (Microsoft)

Category	IT/Corporate / Credential Theft
Impersonates	Microsoft Account Team
Sender	no-reply@access-accsecurity[.]com
Real Domain	microsoft.com
Sending Server	thcultarfdes[.]co.uk (89.144.44.2)
Reply-To	sotrecognizd@gmail.com (attacker's Gmail)
Objective	Trick victim into emailing the attacker directly by disguising contact as account reporting

RISK LEVEL	■ HIGH

Header Analysis — Google Admin Toolbox Result

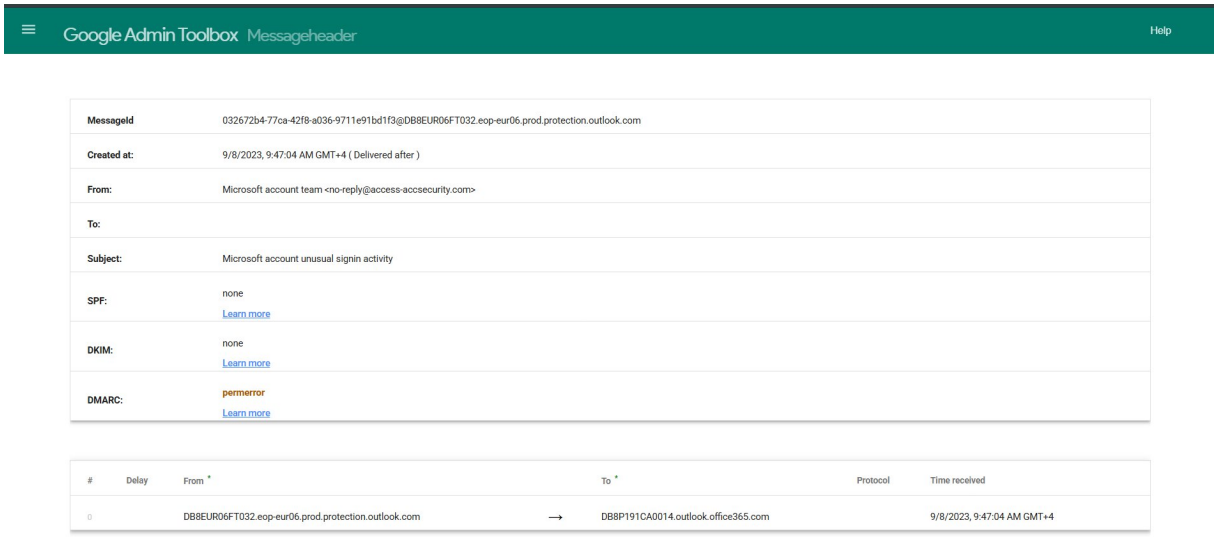

Figure 4: Google Admin Toolbox — Microsoft phishing email: SPF: none, DKIM: none, DMARC: permerror

Red Flags Identified

■ **Fake Sender Domain:** no-reply@access-accsecurity[.]com — Not a Microsoft domain. MXToolbox confirmed this domain has no DNS records whatsoever.

■ **SPF: None:** No SPF record. Microsoft's real emails always pass SPF authentication.

■ **DKIM: None:** Email was not digitally signed — mandatory for all legitimate Microsoft communications.

■ **DMARC: Permerror:** Permanent error meaning the domain has a misconfigured or non-existent DMARC policy.

■ **Reply-To Mismatch:** 'Report The User' button sends email directly to sotrecognizd@gmail.com — the attacker's Gmail, not Microsoft.

■ **Fake Russian Login:** Claims sign-in from Russia/Moscow to maximise fear and urgency.

■ **Tracking Pixel:** Contains a 1×1 invisible tracking image to confirm the email was opened — used to identify active victims.

Attack Flow

Victim receives fake Microsoft security alert about Russian login → Panics → Clicks 'Report The User' → Opens pre-filled email to attacker's Gmail → Attacker conducts further social engineering → Steals Microsoft/Office 365 credentials → Corporate data breach.

4.5 Sample 5 — Banking Fraud (Royal Bank of Canada)

Category	Banking Fraud / Advance Fee Scam
Impersonates	Royal Bank of Canada (RBC)
Sender	RoyalBanlOfCannada@hotmail[.]com
Real Domain	rbc.com
Sending Server	oblszn57[.]ru — Russian server (REG.RU hosting)
Reply-To	rev.innocent-johnson27@outlook.com
Objective	Trick victim into paying a fake $320 USD activation fee for a non-existent $12 million prize

RISK LEVEL	■ HIGH

Header Analysis — Google Admin Toolbox Result

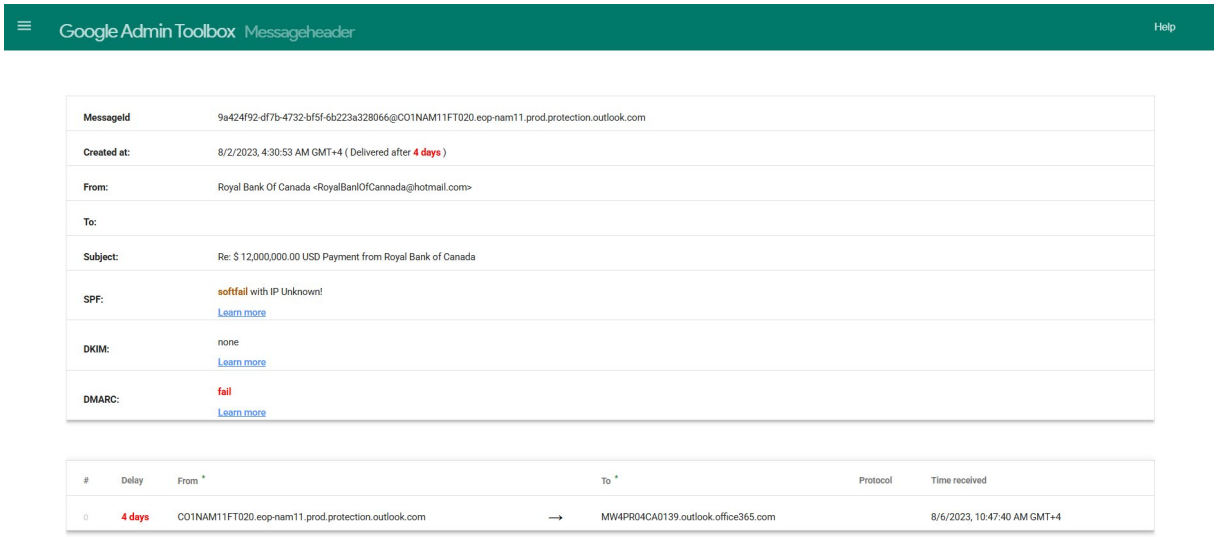

Figure 5: Google Admin Toolbox — Royal Bank: SPF: softfail, DKIM: none, DMARC: fail, 4-day delivery delay

Red Flags Identified

■ **Misspelled Brand:** 'RoyalBanlOfCannada' — both 'Bank' and 'Canada' are misspelled. Legitimate financial institutions never have typos in their email address.

■ **Personal Hotmail:** A real bank with $12 million to transfer would never use a personal Hotmail account.

■ **Russian Server:** Email originated from oblszn57[.]ru — a Russian hosting server. Royal Bank of Canada operates exclusively from Canadian infrastructure.

■ **SPF: Softfail:** Sending IP (62.33.7.21) is not an authorised sender for hotmail.com — email is spoofed.

■ **DMARC: Fail:** Email completely failed DMARC authentication.

■ **4-Day Delivery Delay:** Email was delivered 4 days after being sent — legitimate bank communications are near-instantaneous.

■ **Advance Fee Fraud:** Requests $320 USD activation fee to release $12 million — a classic 419 scam where no prize exists.

■ **Server Blacklisted:** oblszn57[.]ru is listed on UCEPROTECTL3 — an international spam/abuse blacklist.

Attack Flow

Victim receives lottery winning notification claiming $12 million from Coca-Cola/RBC → Excited by prospect of wealth → Sends personal details → Attacker requests $320 activation fee → Victim pays → Attacker disappears → No prize exists — direct financial loss.

5. Domain & Blacklist Investigation

All sender and relay domains identified in the five phishing samples were investigated using MXToolbox SuperTool. The results confirm a consistent pattern of missing or misconfigured email security records across all attacker-controlled domains.

5.1 MXToolbox Domain Lookup Results

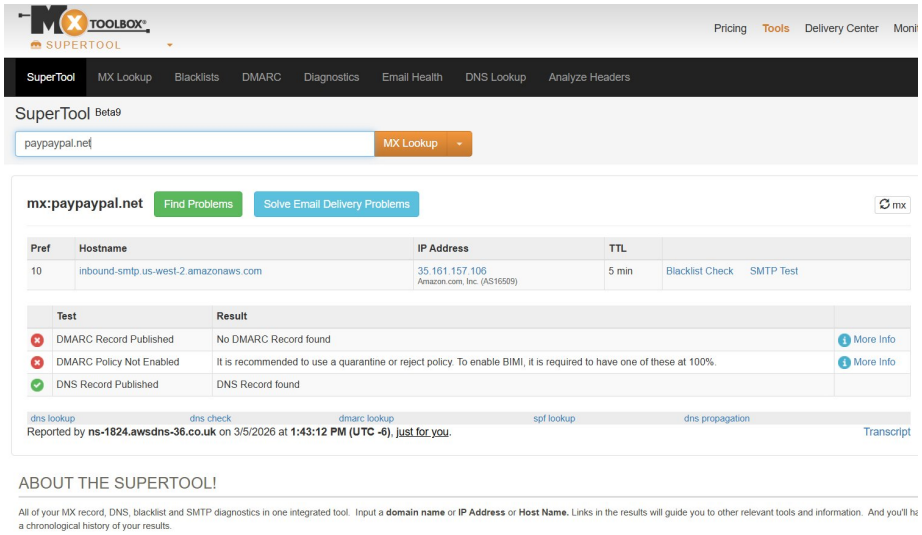

Figure: MXToolbox result for paypaypal.net — No DMARC, no SPF policy. Uses AWS infrastructure unrelated to PayPal.

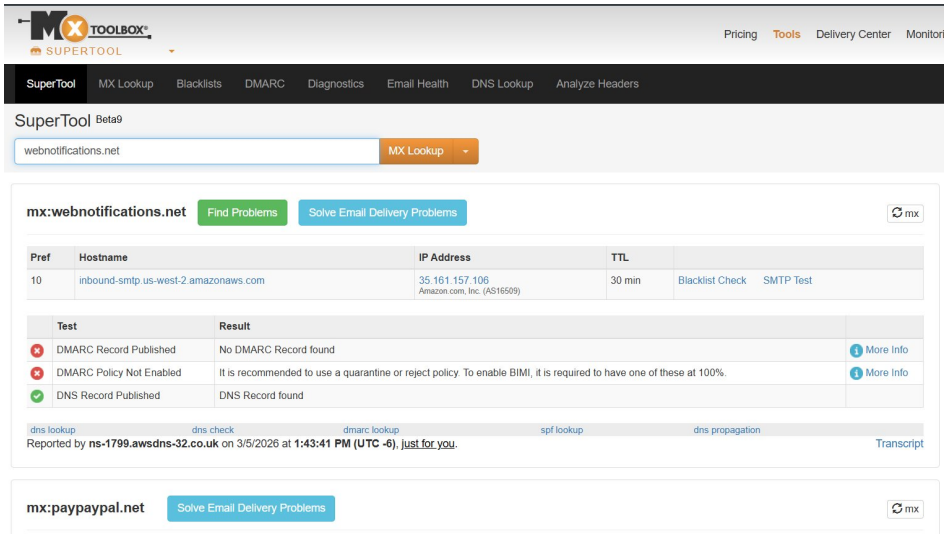

Figure: MXToolbox result for webnotifications.net — Same AWS IP as paypaypal.net, confirming shared attack infrastructure.

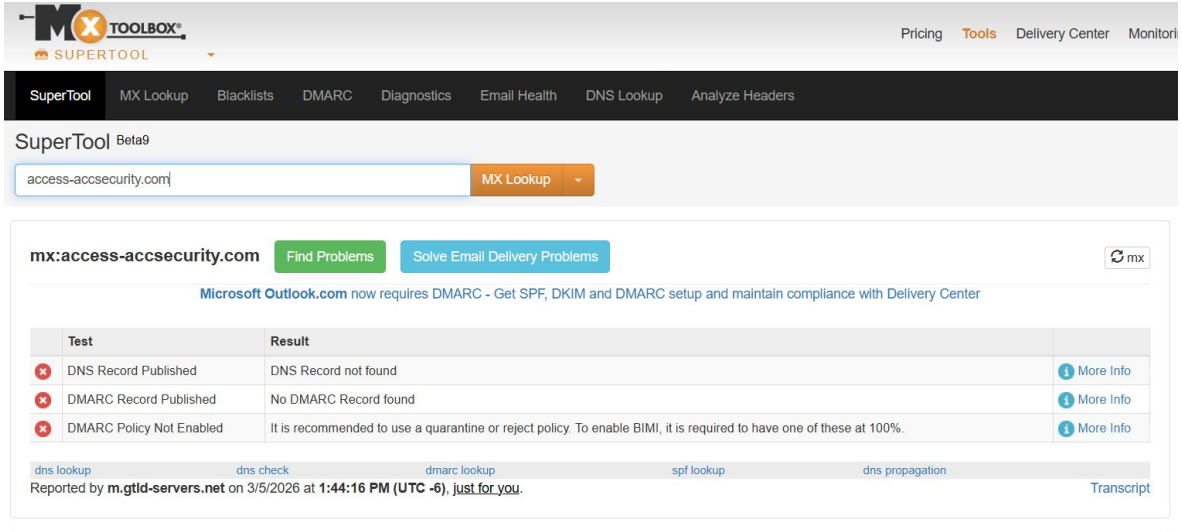

Figure: MXToolbox result for access-accsecurity.com — DNS record NOT FOUND. Domain barely exists online.

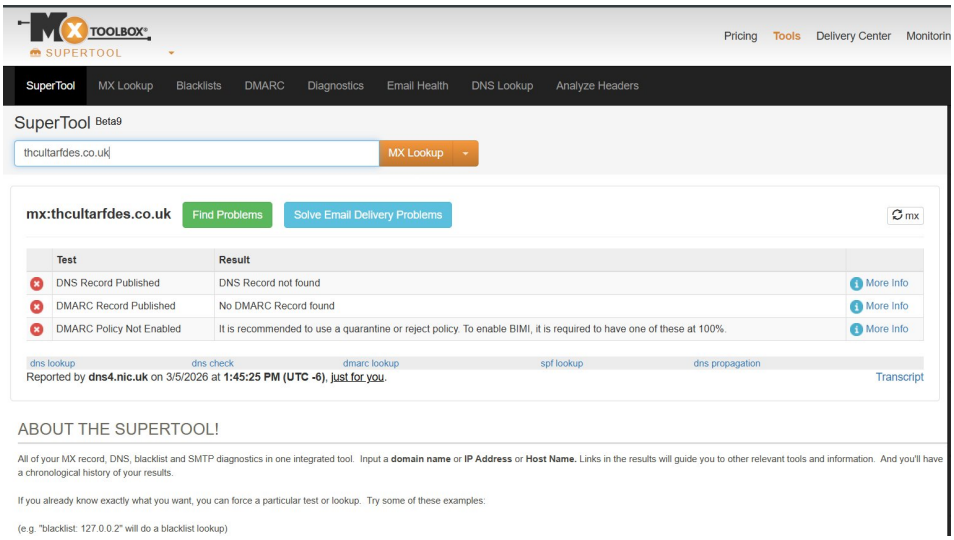

Figure: MXToolbox result for thcultarfdes.co.uk — DNS record NOT FOUND. Ghost domain used only to relay phishing emails.

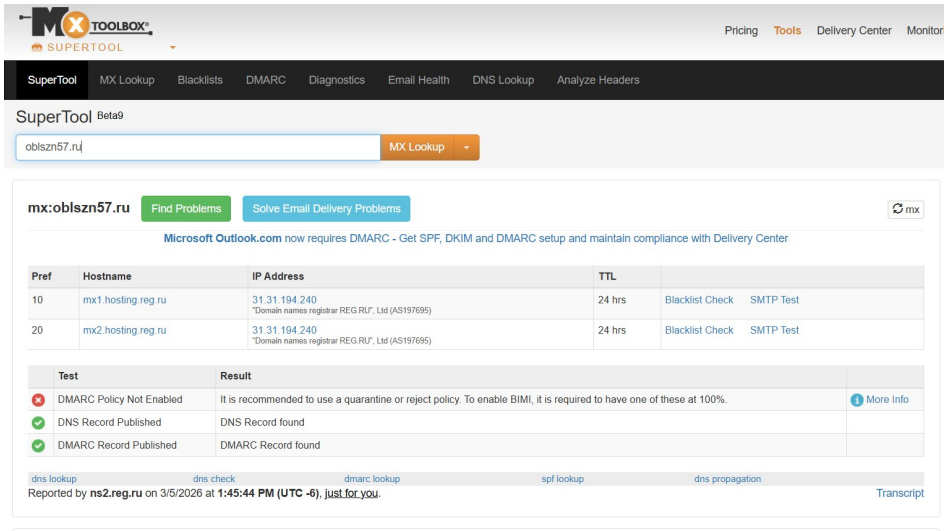

Figure: MXToolbox result for oblszn57.ru — Russian domain (REG.RU registrar). DMARC policy not enabled.

5.2 Blacklist Check Results

Both the sending IP associated with the Russian domain and the Russian domain itself were checked against international spam blacklists. Both were found to be actively listed:

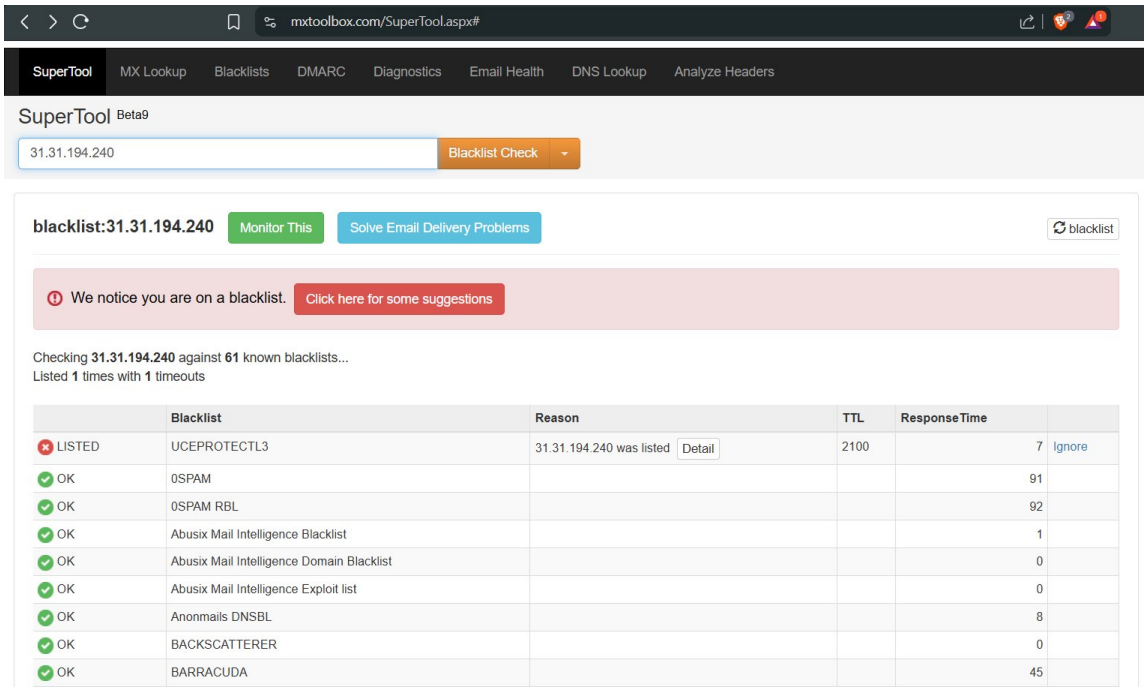

Figure: Blacklist check for IP 31.31.194.240 — LISTED on UCEPROTECTL3 (checked against 61 blacklists)

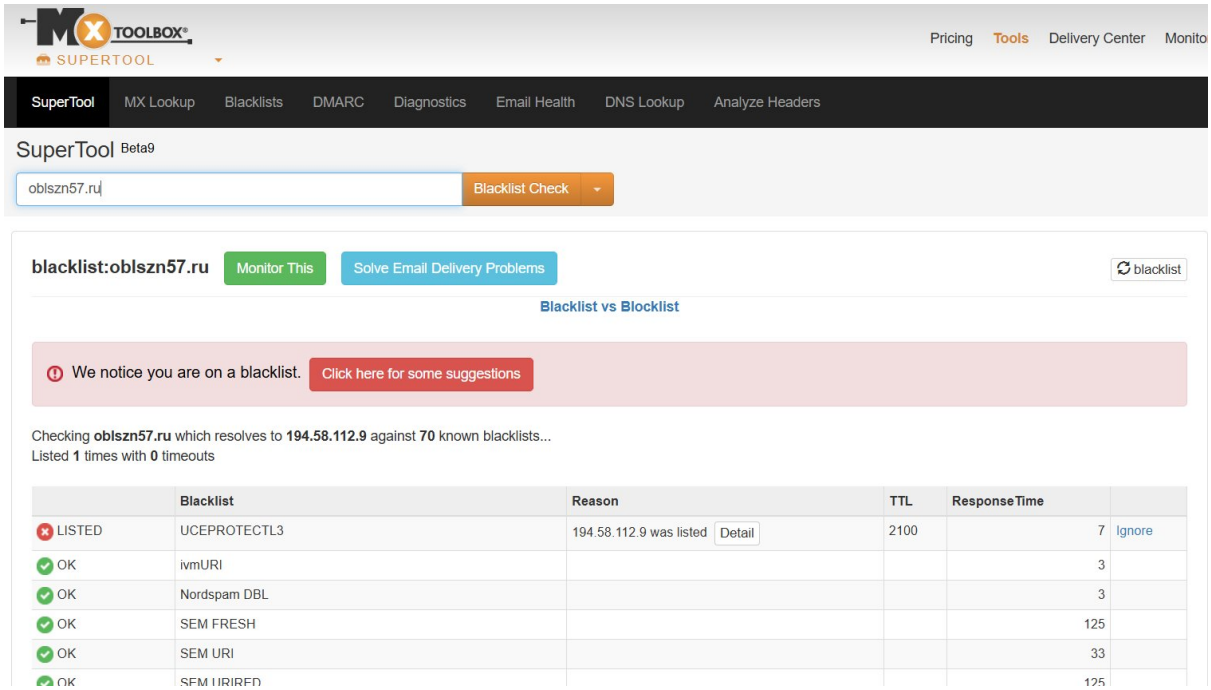

Figure: Blacklist check for oblszn57.ru — LISTED on UCEPROTECTL3 (checked against 70 blacklists)

6. Comparative Analysis & Key Findings

6.1 Authentication Failure Summary

Sample	Impersonates	SPF	DKIM	DMARC	Blacklisted
PayPal	PayPal Inc.	■ None	■ None	■ None	—
Canada Post	Canada Post	■ None	■ None	■ None	—
Instagram	Meta/Instagram	■ None	■ None	■ None	—
Microsoft	Microsoft Corp.	■ None	■ None	■ Permerror	—
Royal Bank	RBC	■ Softfail	■ None	■ Fail	■ YES

6.2 Key Findings

Finding 1 — Shared Attack Infrastructure

Samples 2 (Canada Post) and Sample 3 (Instagram) both used webnotifications[.]net and resolved to the same Amazon AWS IP address (35.161.157.106), confirming a coordinated, multi-brand phishing campaign by a single threat actor.

Finding 2 — Zero Authentication

Not a single phishing email passed any of the three email authentication checks (SPF, DKIM, DMARC) — a definitive technical indicator of email spoofing detectable by properly configured email security gateways.

Finding 3 — Active Blacklist Listings

The sending infrastructure for Sample 5 (Royal Bank) was actively listed on UCEPROTECTL3 — an international spam and abuse blacklist — confirming a documented history of malicious activity.

Finding 4 — Ghost Domains

access-accsecurity[.]com and thcultarfdes[.]co.uk returned no DNS records on MXToolbox — ghost domains created solely to send phishing emails then disappear.

Finding 5 — Multi-Vector Social Engineering

All five samples used psychological manipulation including urgency, fear, financial reward, and impersonation of trusted brands — designed to bypass rational thinking.

7. Risk Classification Summary

#	Sample	Category	Risk	Business Impact
1	PayPal	Financial Fraud	■ HIGH	Company PayPal account drained, direct financial loss
2	Canada Post	Delivery Scam	■ HIGH	Login credentials stolen, account hijacked
3	Instagram	Social Media	■ HIGH	Business account compromised, brand reputation damage
4	Microsoft	Corporate IT Breach	■ HIGH	Office 365 compromise, data breach, business email fraud
5	Royal Bank	Banking Fraud	■ HIGH	Employee sends company funds to fraudster, direct financial loss

8. Prevention Guidelines for Employees

The following guidelines are written in plain, non-technical language for all employees regardless of technical background. These measures directly address the attack techniques observed in the five phishing samples analysed in this report.

8.1 Always Check the Sender's Email Address

Before doing anything with an email, look at the actual email address — not just the display name. Attackers can make the display name say 'PayPal Support' while the actual address is support@paypaypal.net. Always hover over or click the sender name to reveal the real address.

8.2 Do Not Click Links Under Pressure

If an email creates urgency — 'Act now or lose access', 'Your account will be locked' — treat this as a red flag. Legitimate companies give adequate time to respond and will never threaten immediate consequences for not clicking a link.

8.3 Go Directly to the Website

If you receive an alert about your PayPal, bank, Instagram, or Microsoft account, open your browser and type the official website address yourself (e.g. paypal.com). Never use a link inside an email to access your account.

8.4 Check for Grammar and Spelling Errors

Errors like repeated words ('pending parcel pending pickup') or misspelled brand names ('RoyalBanlOfCannada') are strong indicators of phishing. Legitimate organisations employ professional communication teams.

8.5 Verify Unusual Financial Requests Independently

No legitimate bank or company will ask you to pay a fee to receive a prize. If you receive any communication requesting a payment to 'release funds', call the organisation directly using the phone number from their official website — never from the email.

8.6 Enable Multi-Factor Authentication (MFA)

Even if attackers steal your password, MFA prevents them from logging in without a second verification step. Enable MFA on all critical accounts including email, banking, and social media.

8.7 Quick Reference — Do's and Don'ts

■ DO	■ DON'T
Check the actual sender email address	Trust display names alone
Go directly to official websites	Click links inside suspicious emails
Report suspicious emails to IT/security	Ignore or delete without reporting
Enable MFA on all accounts	Use the same password everywhere
Call to verify unexpected financial requests	Send money based on email instructions
Check for spelling/grammar errors	Assume professional design = legitimate

Hover over links before clicking	Click shortened or obfuscated URLs

9. Conclusion

This report analysed five real-world phishing email samples spanning financial fraud, delivery scams, social media account takeover, corporate IT credential theft, and banking fraud. Technical investigation confirmed that every single sample failed all three email authentication standards (SPF, DKIM, DMARC), with two sending servers actively listed on international spam blacklists.

A particularly significant finding was the identification of shared attack infrastructure between the Canada Post and Instagram phishing samples — both using the same fake domain and the same IP address — confirming a coordinated multi-brand campaign by a single threat actor.

Organisations that implement the prevention guidelines outlined in Section 8, combined with properly configured email authentication and Multi-Factor Authentication on all critical accounts, will significantly reduce their exposure to phishing attacks of this nature.

10. References

[1]	Google Admin Toolbox — Message Header Analyzer	toolbox.googleapps.com/apps/messageheader/
[2]	MXToolbox SuperTool — Domain & Blacklist Analysis	mxtoolbox.com/SuperTool.aspx
[3]	rf-peixoto/phishing_pot — Real Phishing Samples	github.com/rf-peixoto/phishing_pot
[4]	OWASP Phishing Prevention Cheat Sheet	cheatsheetseries.owasp.org
[5]	Future Interns — Cyber Security Task 2 Brief	linkedin.com/company/future-interns
[6]	caniphish.com — Phishing Email Templates (Educational)	caniphish.com/free-phishing-test/phishing-email-templates

This report was prepared by Mohammad Moinuddin (CIN ID: FIT/FEB26/CS6460) as part of the Future Interns Cyber Security Internship Programme 2026. All analysis was conducted ethically and within the defined read-only scope. No systems were compromised or harmed during this assessment.